

- [ABR Physics Help](#)
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$$ICF = \frac{\text{Dose in inhomogeneous medium}}{\text{Dose in homogeneous medium}}$$

1. Method: Effective attenuation coefficient

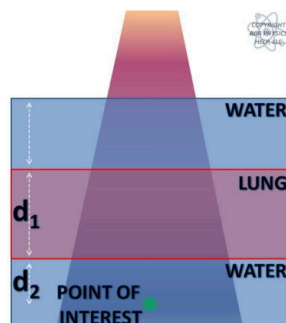
- $ICF = e^{\mu(d-d')}$
- μ = the effective attenuation coefficient of water for a beam quality.
- d = the physical depth from the surface to the point of calculation.
- d' = the equivalent depth weighted by relative electron densities.

2. Method: Ratio of Tissue-Air Ratios (RTAR)

- $ICF = \frac{TAR(d', W)}{TAR(d, W)}$
- W = the field size at the level of the point of interest.

3. Method: Power Law (Batho)

- $ICF = \frac{TAR(d_1, W)^{\rho_1 - \rho_2}}{TAR(d_2, W)^{\rho_1 - \rho_2}}$
- ρ_1 = the relative electron density of the medium in which the point of calculation lies.
- ρ_2 = the relative electron density of the overlying material.
- d_1 = the depth within this medium.
- d_2 = the distance to the upper surface of the overlying inhomogeneity slab.
- Note that the TAR ratios can be replaced by TMR or TPR.



4. Method: Equivalent tissue-air ratio (ETAR)

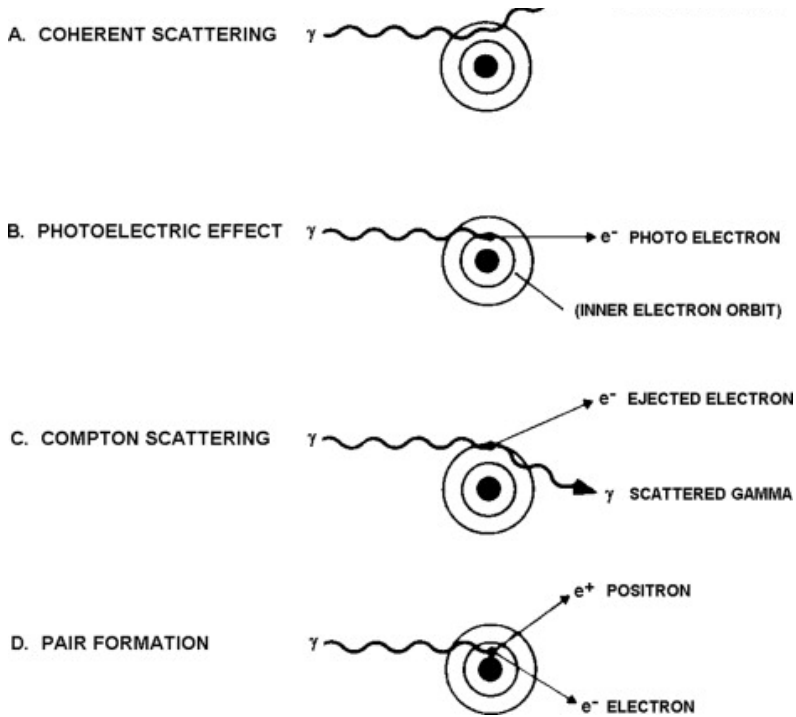
- This method is also based on a ratio of TAR values and is termed effective because it is scaled to an effective depth and field size.

5. Method: Superposition/Convolution

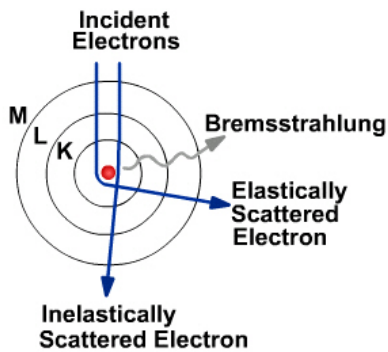
- This method consists of two parts: a TERMA part which describes the distribution of the energy that is released at a primary interaction site and is available for consumption in the medium and a kernel part that describes the distribution of the scatter from a primary interaction (that is, the kernel dictates how the TERMA is going to be distributed in the medium).

Review of Particle Interactions

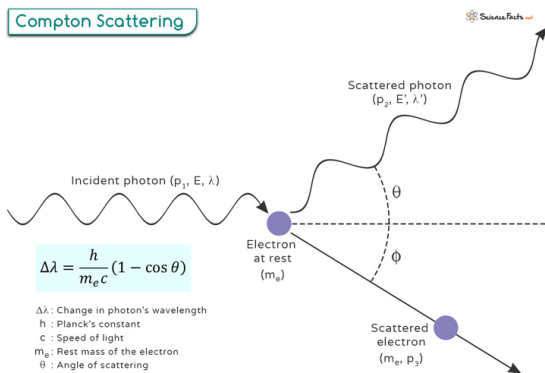




Bremsstrahlung

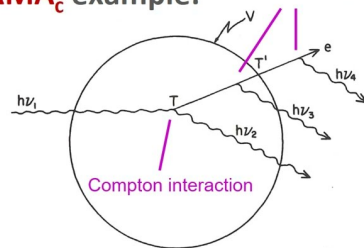


Compton Scattering



TERMA, KERMA, & KERMA_c example: bremsstrahlung interactions

V = volume of interest
 $h\nu_1$ = original photon
 e = Compton electron
 T = Kinetic energy of e
 $h\nu_2$ = Compton photon
 $h\nu_3$ = bremsstrahlung photon 1
 $h\nu_4$ = bremsstrahlung photon 2
 T' = residual kinetic energy of e after leaving volume V



$$\text{TERMA} = T + h\nu_2 = h\nu_1$$

$$\text{KERMA} = T$$

$$\text{KERMA}_c = T - h\nu_3 - h\nu_4$$

$$\text{KERMA}_R = h\nu_3 + h\nu_4$$

$$\text{Dose to volume } V = T - h\nu_3 - T'$$

Attix, Introduction to Radiological Physics and Radiation Dosimetry Wiley-VCH, 1st edition (January 8, 1991)

TERMA - Mass Attenuation Coefficient (LOCAL energy deposition, no electron transport).

- Kinetic energy of e^- + Compton Scattered Photons (Photon $\rightarrow$ Photon + Ejected Electron).
- Includes energy deposits that are not transferred to the medium (Compton Scattered Photons).

KERMA - Mass Energy Transfer Coefficient

- Kinetic energy of e^- only (Included Bremsstrahlung).
- DOES NOT INCLUDE COMPTON SCATTERED PHOTONS

Collisional KERMA - Mass Energy Absorption Coefficient

- Kinetic energy of e^- EXCEPT Bremsstrahlung (not a local deposit)
- DOES NOT INCLUDE BREMSSTRAHLUNG OR COMPTON

Radiative KERMA - Fraction of electron energy lost through Bremsstrahlung photon

- BREMSSTRAHLUNG component of KERMA (Incident Electron near the nucleus $\rightarrow$ Change of direction Electron + Ejected Photon).

Dose = TERMA (Kinetic energy of e^- + Compton)

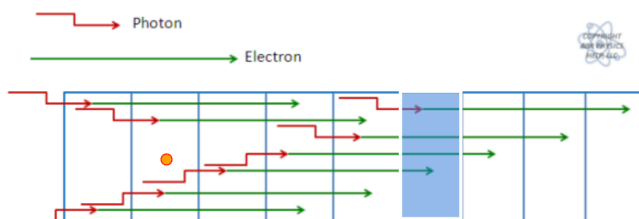
- Radiative KERMA Bremsstrahlung (distantly deposited photons)
- Residual kinetic energy of e^- after leaving volume

Dose = Collisional KERMA - Residual energy after e^- leave volumes (Pays attention to GLOBAL energy deposition using Electron Transport).

Question: Absorbed Dose = Collisional KERMA when there is no Bremsstrahlung or Compton?

Question 3

Please click on the voxel where the absorbed dose is equal to the collision KERMA.



	TERMA	DOSE
1D	<i>Local energy deposition (No electron transport)</i> Category 1 1.1 Linear attenuation 1.2 Ratio of TAR (RTAR) (Equivalent path length, effective SSD, isodose shift) 1.3 Power law (Batho)	<i>Non-local energy deposition (Electron transport)</i> Category 3 3.1 Convolution (pencil beam) 3.2 FFT techniques
3D	Category 2 2.1 Equivalent TAR (ETAR) 2.2 Differential SAR (DSAR) 2.3 Delta volume (DVOL) 2.5 3D Beam Subtraction Method	Category 4 4.1 Superposition/Convolution 4.2 Monte Carlo 2.4 Differential TAR (dTAR)